

Appointments

according to the relative importance of the stratum in the population.

The sample which is the aggregated of the sample units of each of the stratum is known as stratified sampling.

$\bar{x}$, σ^2 : sample mean.

$$S^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$
 : sample variance.

$$\bar{x} \sim N(\mu, \frac{\sigma^2}{n})$$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$
 σ is known.

$$\frac{nS^2}{\sigma^2} \sim \chi^2_{n-1}$$

Remarks

If σ unknown

then
$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

If $E(\text{statistic}) = \text{parameter}$
then statistic is unbiased

2001
Ex.

Friday

16

MAR

The standard deviation of a population is 2.70 inches. Find the probability

that a random sample of size 66

(i) the sample mean will differ from population mean by ~~0.07~~ 0.75 inch or more (ii) the sample mean will exceed the population mean by 0.75 inch or more

Given $P(0 \leq Z \leq 2.25) = 0.4875$

fn. $n = 66$, $\sigma = 2.70$

$$\bar{x} \sim N(\mu, \sigma/\sqrt{n})$$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

$$\begin{aligned} \textcircled{1} P[|\bar{x} - \mu| \geq 0.75] &= 1 - P[|\bar{x} - \mu| < 0.75] \\ &= 1 - P\left[\left|\frac{\sigma}{\sqrt{n}} Z\right| < 0.75\right] \end{aligned}$$

$$= 1 - P\left[|Z| < 0.75 \frac{\sqrt{n}}{\sigma}\right]$$

$$= 1 - 2P\left[0 < Z < 0.75 \frac{\sqrt{n}}{\sigma}\right]$$

$$= 1 - 2P\left[0 < Z < 0.75 \frac{\sqrt{66}}{2.7}\right]$$

$$= 1 - 2P[0 < Z < 2.25] = 0.0246$$

Remarks

APRIL

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Appointments

$$\begin{aligned} \text{ii) } P[\bar{X} - \mu > 0.75] &= P[Z > 0.75 \sqrt{n}] \\ &= P(Z > 2.25) = 0.5 - P(0 < Z < 2.25) \\ &= 0.5 - 0.4877 = 0.0123. \end{aligned}$$

Ex. A normal population has a mean of 0.1 and standard deviation of 2.1. Find the ~~mean~~ probability that mean of a sample of size 900 will be negative.

$$X \sim N(\mu, \sigma^2), \bar{X} \sim N(\mu, \frac{\sigma^2}{n})$$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

$$\Rightarrow \frac{\bar{X} - 0.1}{2.1/\sqrt{900}} = \frac{\bar{X} - 0.1}{2.1/30}$$

$$= \frac{\bar{X} - 0.1}{0.07}$$

$$\Rightarrow \bar{X} = 0.1 + 0.07Z$$

$$\begin{aligned} \therefore P(\bar{X} < 0) &= P(0.1 + 0.07Z < 0) \\ &= P\left(Z < -\frac{0.1}{0.07}\right) = P(Z < -1.43) \\ &= P(Z > 1.43) \end{aligned}$$

MARCH

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

MAR.SUN.18

$$= 0.5 - P(0 \leq Z \leq 1.43) = 0.5 - 0.4236 = 0.0764.$$

Appointments

Ex: The guaranteed average life of ~
certain types of electric light ^{bulbs} is
1000 hrs. with S.D. 125 hrs. It is
decided to sample the output so as
to ensure that 90% of the light ^{phones to make}
bulbs do not fall ~~short~~ short of
guaranteed average by more than
2.5%. What is the minimum size of
the sample?

phones to make

gk. $\mu = 1000, \sigma = 125.$

$\pi \times 1000 - 2.5\% \text{ of } 1000 \times \pi \times 1000 - 25$ Remarks
 $= 975.$

Remarks

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

$$\therefore Z = \frac{\pi - 100.0}{125/\sqrt{n}} = \frac{975 - 1000}{125/\sqrt{n}} = \frac{\sqrt{n}}{5}$$

now $P(Z > \sqrt{n}/5) = 0.90$

$$\Rightarrow P(0 < Z < \sqrt{n}/5) = 40$$

2001

Tuesday

$$\Rightarrow \frac{\sqrt{h}}{5} = 1.28 \Rightarrow h = 25 (1.28)^2 \approx 41.$$

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